Assignment Set 2

CS-423 (Artificial Intelligence)

PART -V (Total Marks 2*10=20)

1) Hill climbing can get stuck due to which of the following phenomenon?

- [x] Local Maxima
- [ ] Inadmissible heuristic
- [x] Ridges
- [x] Plateau

2) Which of the following will eventually find the optimal solution?

- [x] Random Walk
- [ ] Greedy Hill Climbing
- [x] Hill Climbing with random restarts
- [ ] Local Beam Search

3) What happens to the probability of making a move that leads to a state worse than the current state as we reduce the temperature in simulated annealing?

- ( ) Increases
- (✓) Decreases
- ( ) Remains the same

4) Suppose a hill-climbing procedure has a probability p=0.2 of getting successful in a particular run. What is the expected number of runs required for hill-climbing with random restarts to be successful?

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5) Genetic algorithms are said to jump from one hill to another. Which of the following is responsible for such behavior?

- ○ Mutation
- ✓ Cross over
- ○ Fitness function
- ○ Natural Selection

6) Which of the following pair(s) of algorithms is/are equivalent?

- ☐ Tabu Search, Greedy Hill-Climbing with Sideways moves
- ✓ Simulated Annealing with infinite temperature, Random Walk.
- ☐ Local Beam Search with infinite beam size, Breadth-First Search.
- ☐ Genetic Algorithms with unit population size, Random Sampling

7) Find the no. of steps taken by the Greedy Hill-Climbing Search to go from State A to State B where h(s) = no. of misplaced tiles in state S w.r.t state B.

A

3	1	2
4	5	8
6		7

B

	1	2
3	4	5
6	7	8

5

8) Which of the following statement(s) is/are correct about Local Beam Search?

- ☐ It will not find the optimal solution for a beam size of k if it does not find the optimal solution for some beam size > k.
- ✓ It might find the optimal solution for a beam size of k even if it does not find the optimal solution for some beam size > k.
- ✓ It has a better chance of finding the optimal solution for higher values of k.
- ✓ It is the same as Greedy Hill-Climbing with k = 1.

9) What is the direction of steepest descent to find the minimum of the function $f(x, y, z)= e^x(xy + 2z)$ at (x, y, z) = (0, 1, -1).

- ○ (2, -1, 0)
- ○ (2, 0, -2)
- ○ (-1, -1, 0)
- ✓ (1, 0, -2)

10) Which of the following statement(s) is/are not correct about Tabu Search?

- ✓ It never visits the same state more than once.
- ☐ It uses past experience to explore unvisited areas of the search space.
- ✓ The transition from global to local optimization is marked by increasing the length of tabu list.
- ✓ It always escapes a local optimum.